



Human-Robot Interaction, Device Prototyping, and Embodied AI

Huaishu Peng | UMD CS | Fall 2025

It's a **brand-new** course (and like our first prototype)

- not everything will be perfect
- but we will try!

TA: Zeyu Yan

zeyuy@umd.edu

PhD Student, CS

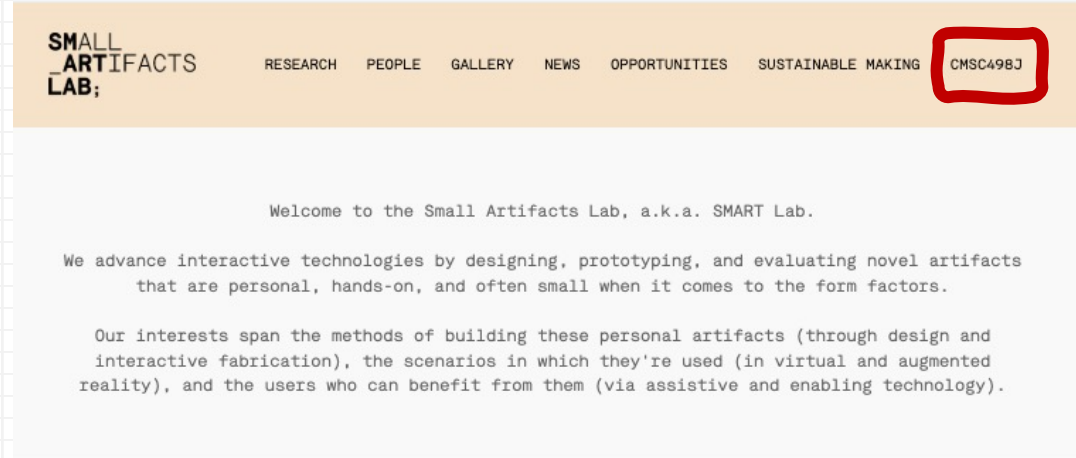
Zeyu Yan is a CS PhD student with a mechanical engineering background.

He works in the area of Human Computer Interaction (HCI) within the field of Digital Fabrication, Tangible User Interface, Accessibility, and Haptic Interaction. His recent work revolves around digital fabrication techniques that supports sustainable printed circuit board (PCB) prototyping.



Huaishu Peng

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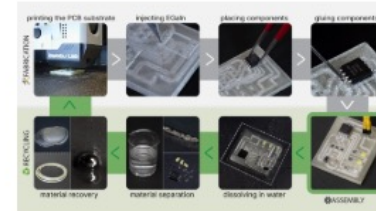
Publications

UIST 2025 | UIST Best Paper Award **NEW**

DissolvPCB: Fully Recyclable 3D-Printed Electronics with Liquid Metal Conductors and PVA Substrates

Zeyu Yan, SuHwan Hong, Josiah Hester, Tingyu Cheng, Huaishu Peng

[DOI](#) [PDF](#) [GITHUB](#)



CSCW 2025 **NEW**

Verisimilitude as Boon and Bane: How People Initiate Opportunistic Interactions at Professional Events in Social VR

Victoria Chang, Caro Williams-Pierce, Huaishu Peng, Ge Gao

[DOI](#) [PDF](#)

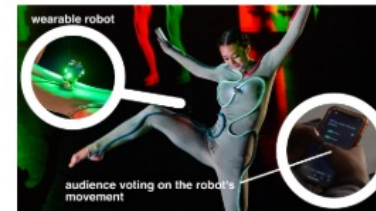


DIS 2025 **NEW**

Cybernetic Marionette: Channeling Collective Agency Through a Wearable Robot in a Live Dancer-Robot Duet

Anup Sathya, Jiasheng Li, Zeyu Yan, Adriane Fang, Bill Kules, Jonathan David Martin, Huaishu Peng

[DOI](#) [PDF](#) [GITHUB](#)

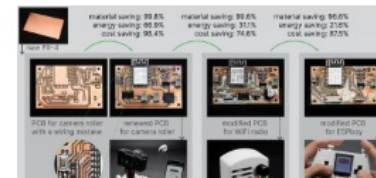


CHI 2025 | CHI Best Paper Honorable Mentions Award **NEW**

PCB Renewal: Iterative Reuse of PCB Substrates for Sustainable Electronic Making

Zeyu Yan, Advait Vartak, Jiasheng Li, Zining Zhang, Huaishu Peng

[DOI](#) [PDF](#) [GITHUB](#)



Small Artifacts Lab

<https://smartlab.cs.umd.edu/>

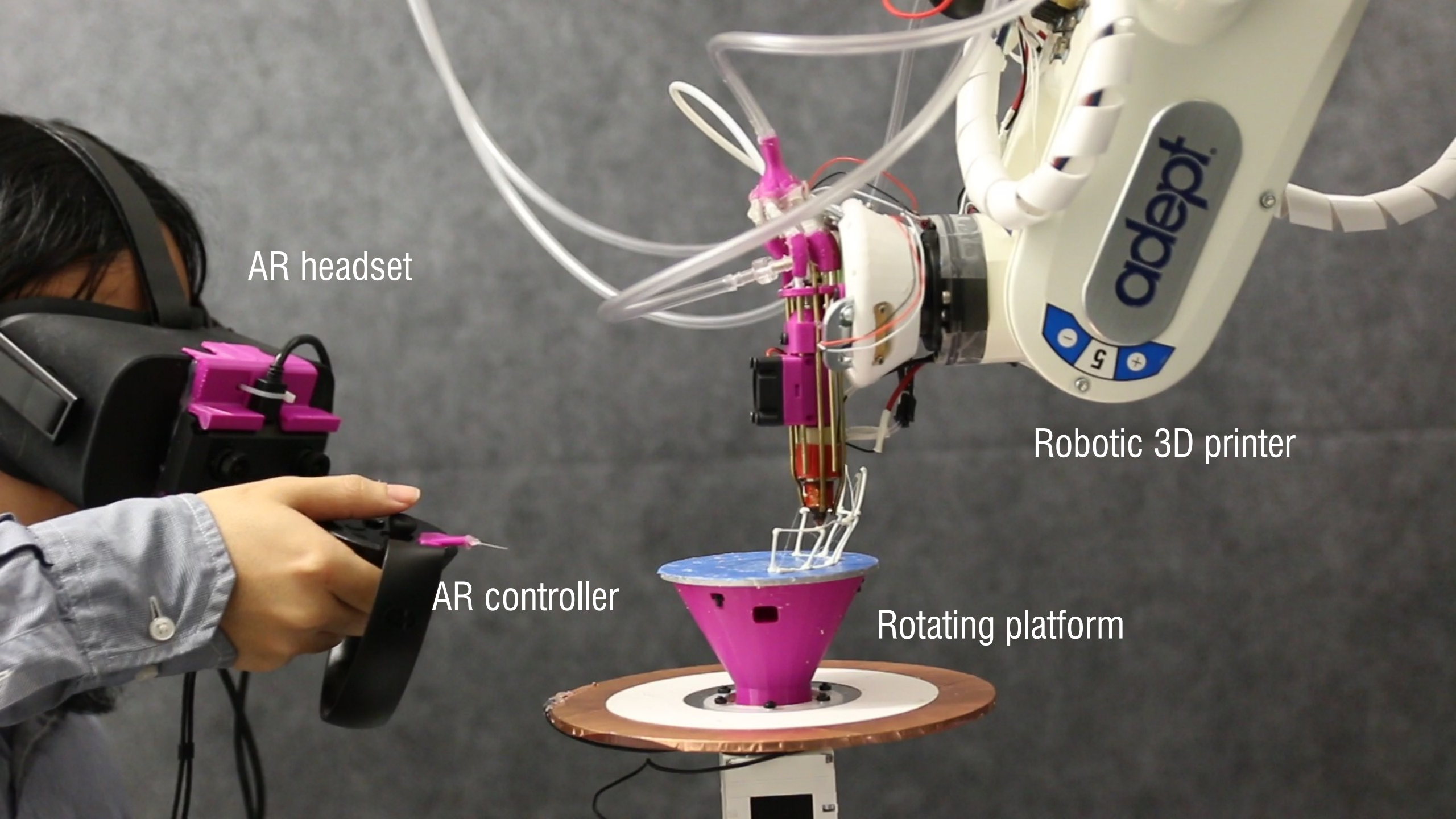
RoMA: Interactive Fabrication with Augmented Reality and a Robotic 3D Printer

Huaishu Peng | Jimmy Briggs | Cheng-Yao Wang | Kevin Guo | Joseph Kider | Stefanie Mueller | Patrick Baudisch | François Guimbretière



Cornell University



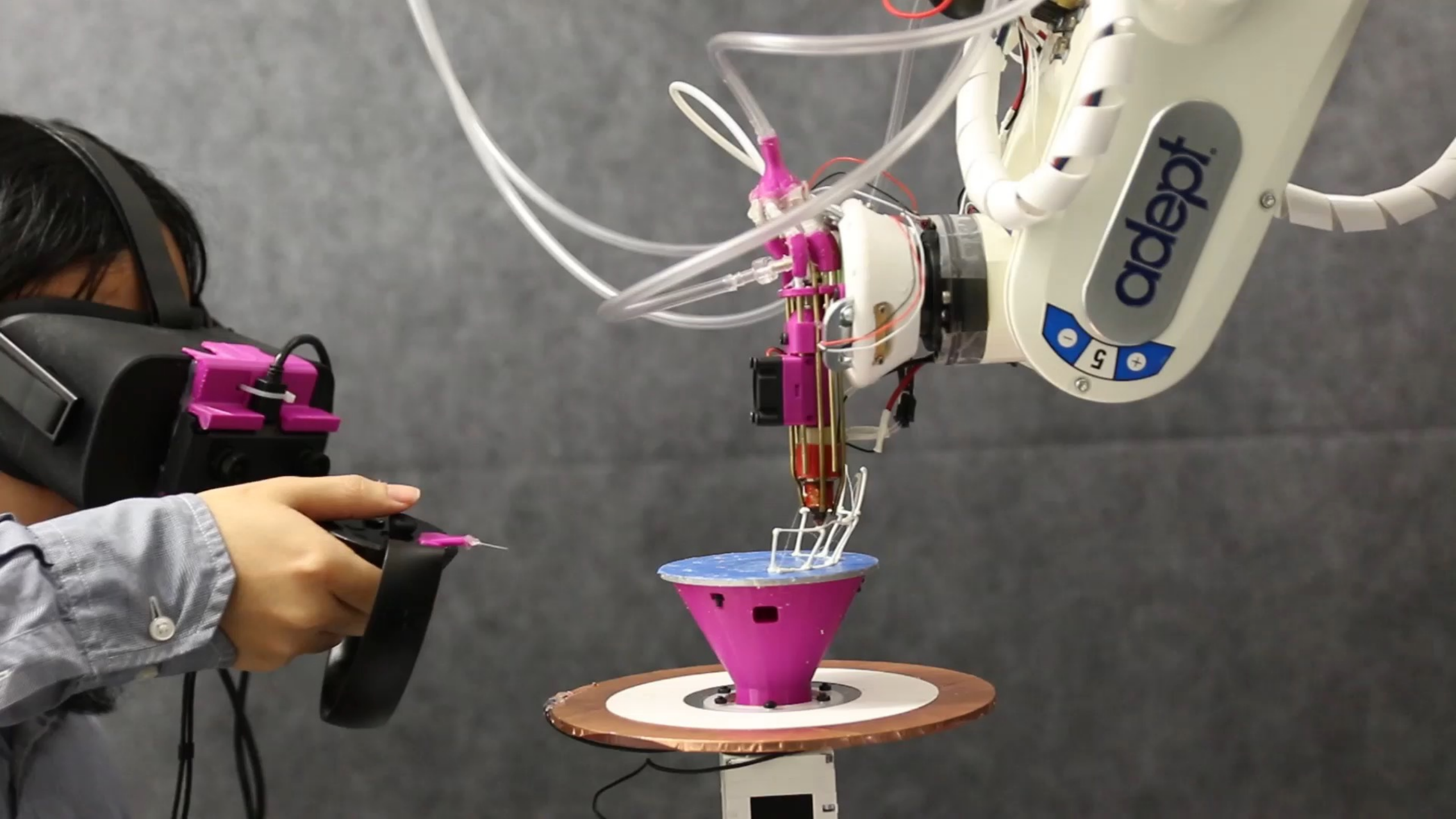


AR headset

Robotic 3D printer

AR controller

Rotating platform

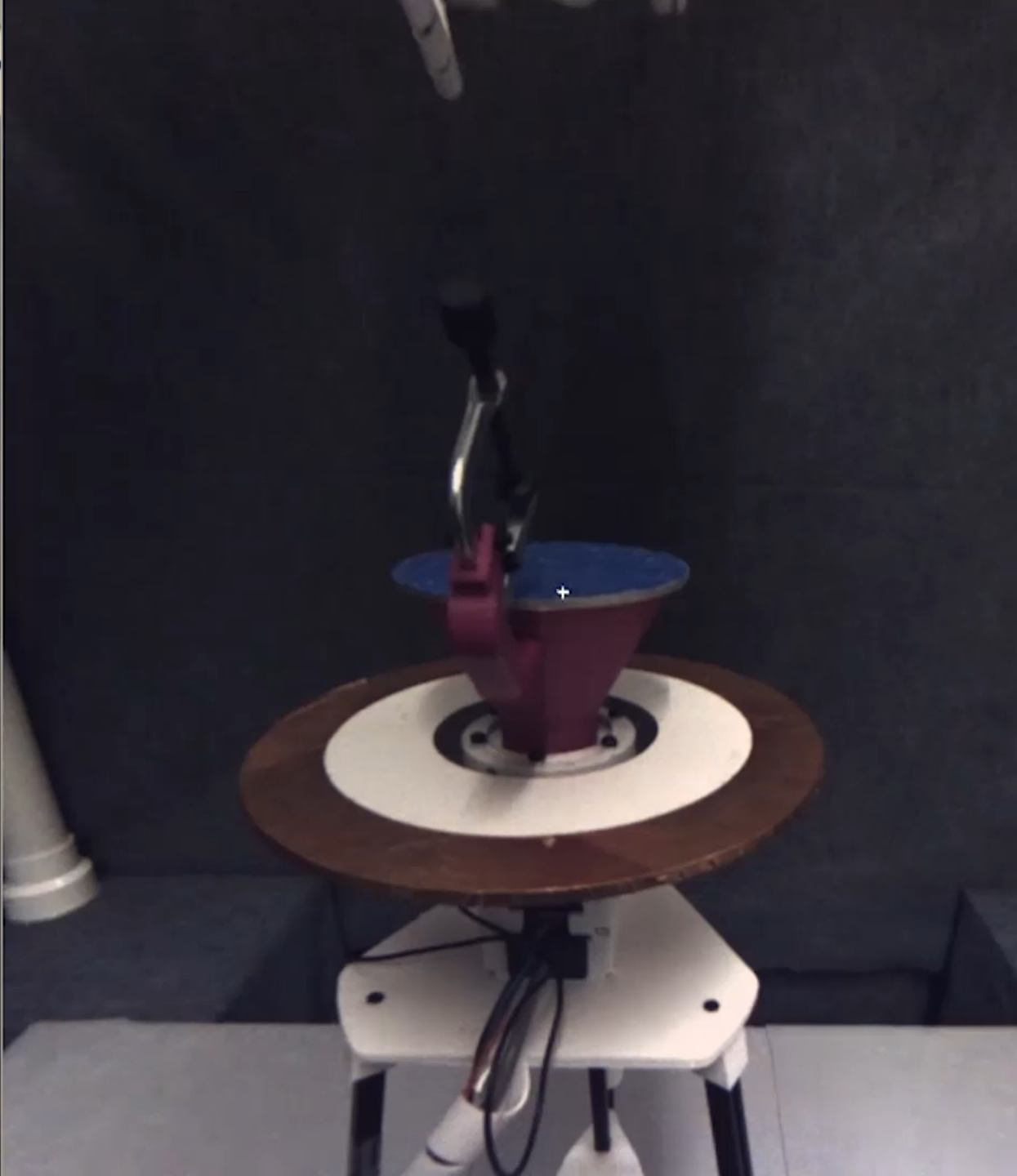


X3



The designer creates a teapot body with *Revolve*

X5



Not all robots have arms and not all robots look like robots.



Cybernetic Marionette:

Channeling Collective Agency Through a Wearable Robot in a Live Dancer-Robot Duet

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SMALL
ARTIFACTS
LAB;

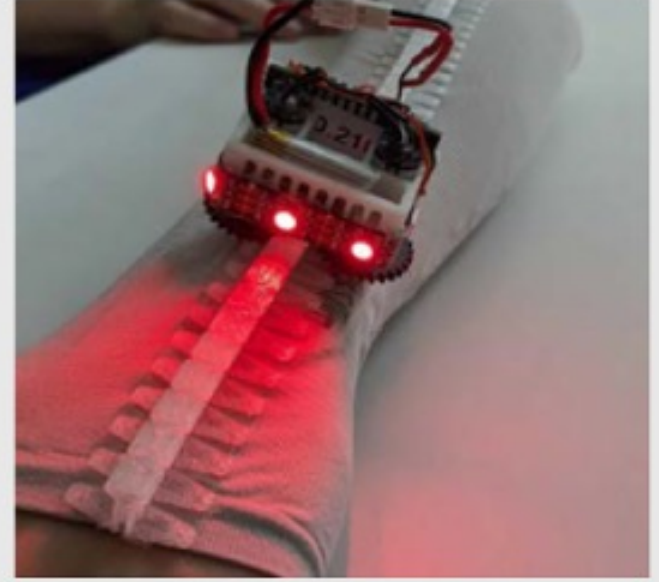
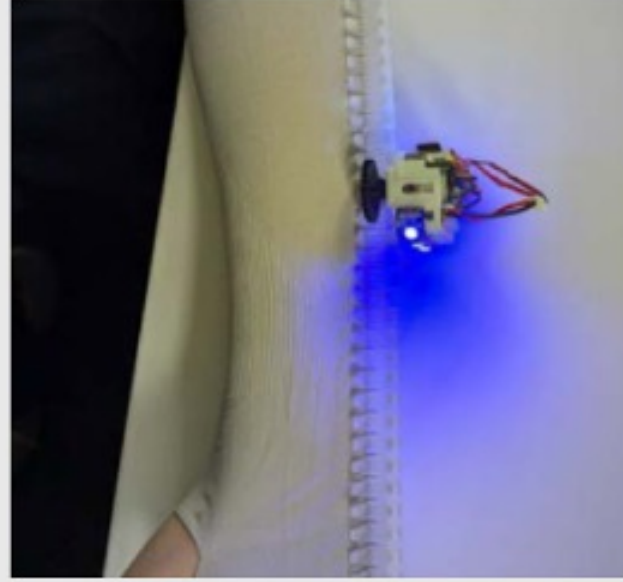
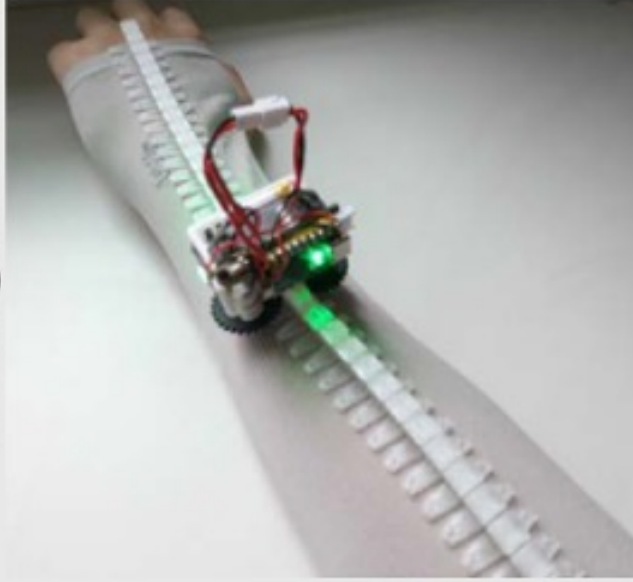
The Design of On-Body Robots for Older Adults

HRI 2025

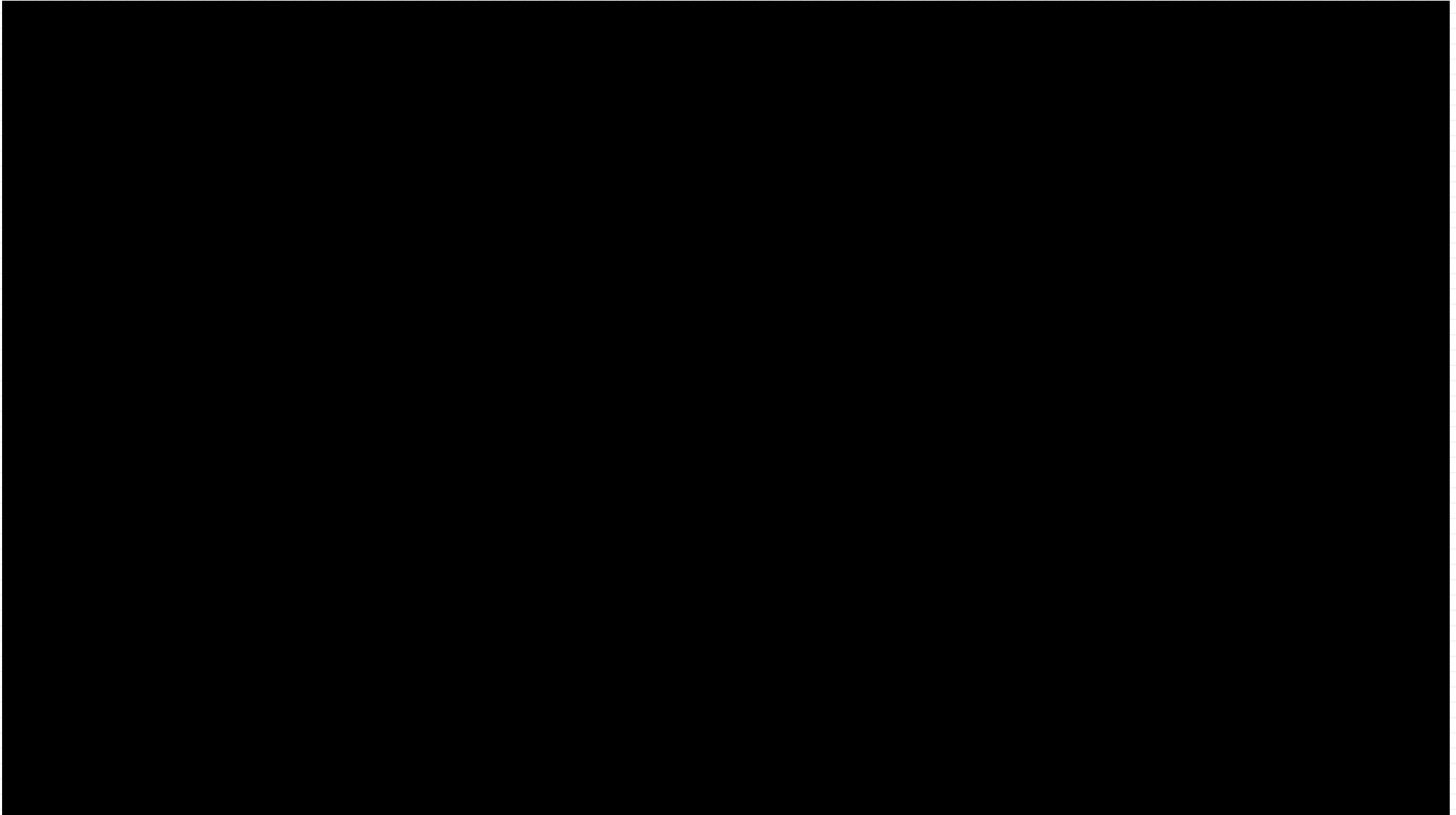
Co-design
On-body Robots
with Older Adults



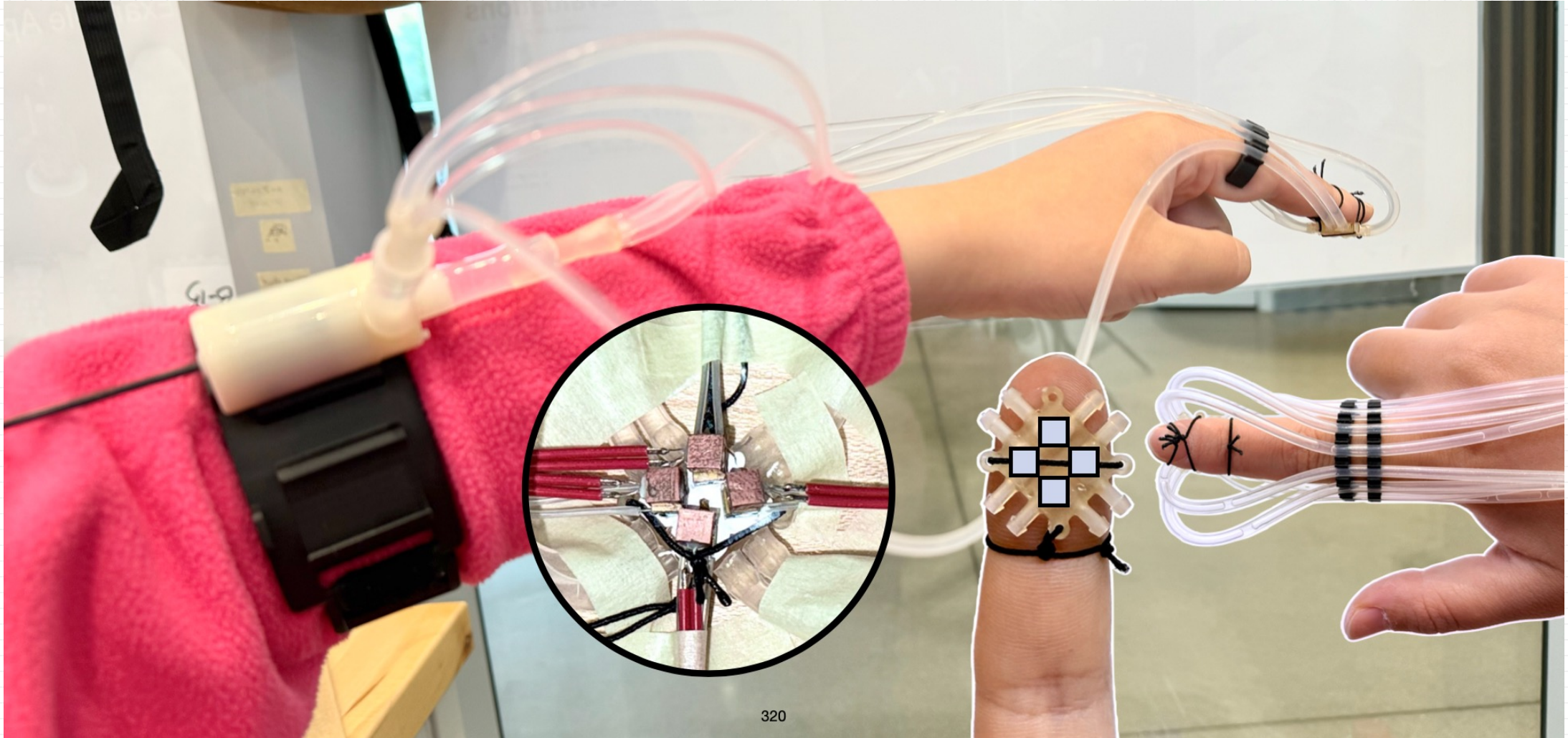
Design Probe



Adding haptics

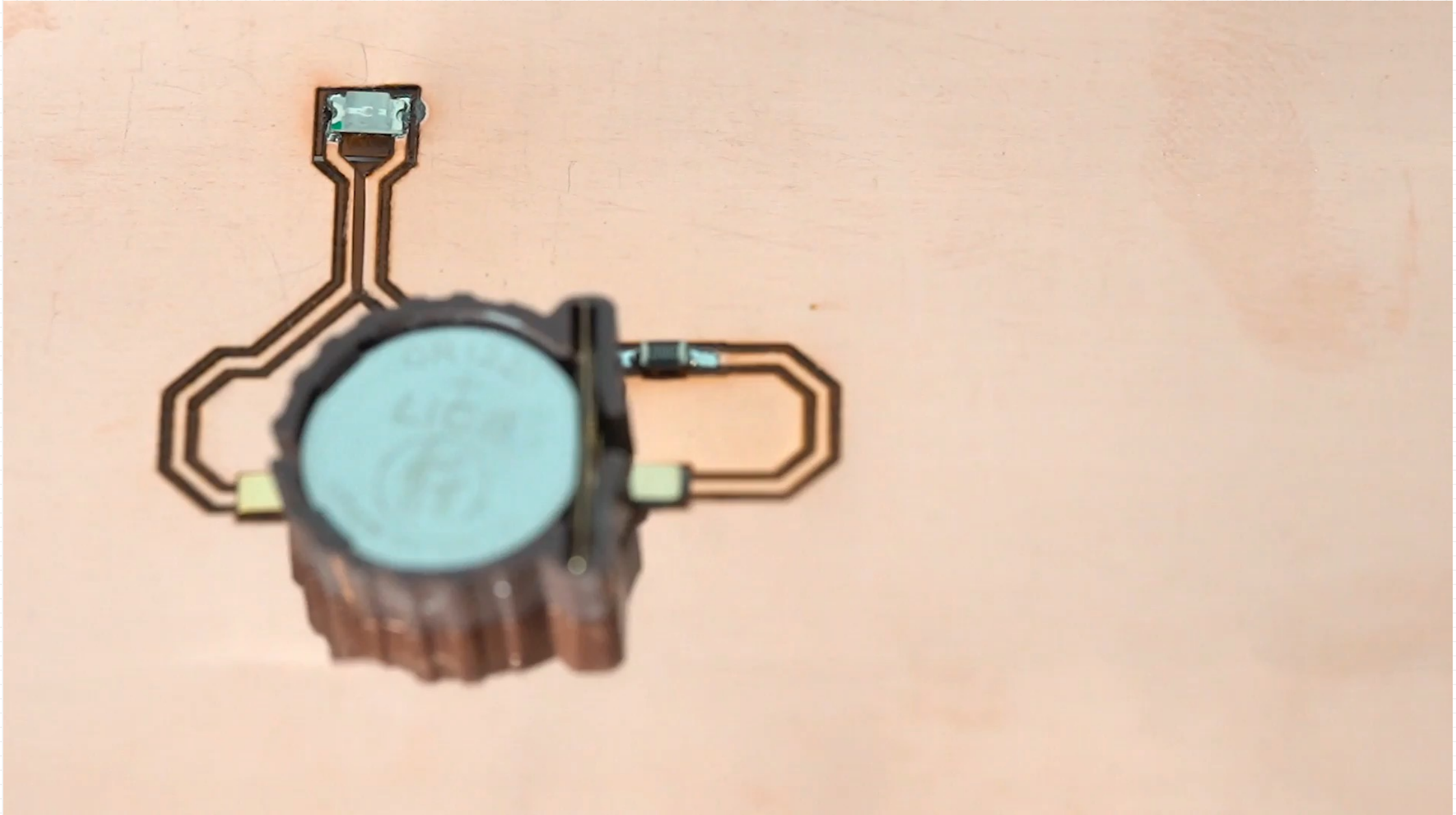


Adding haptics

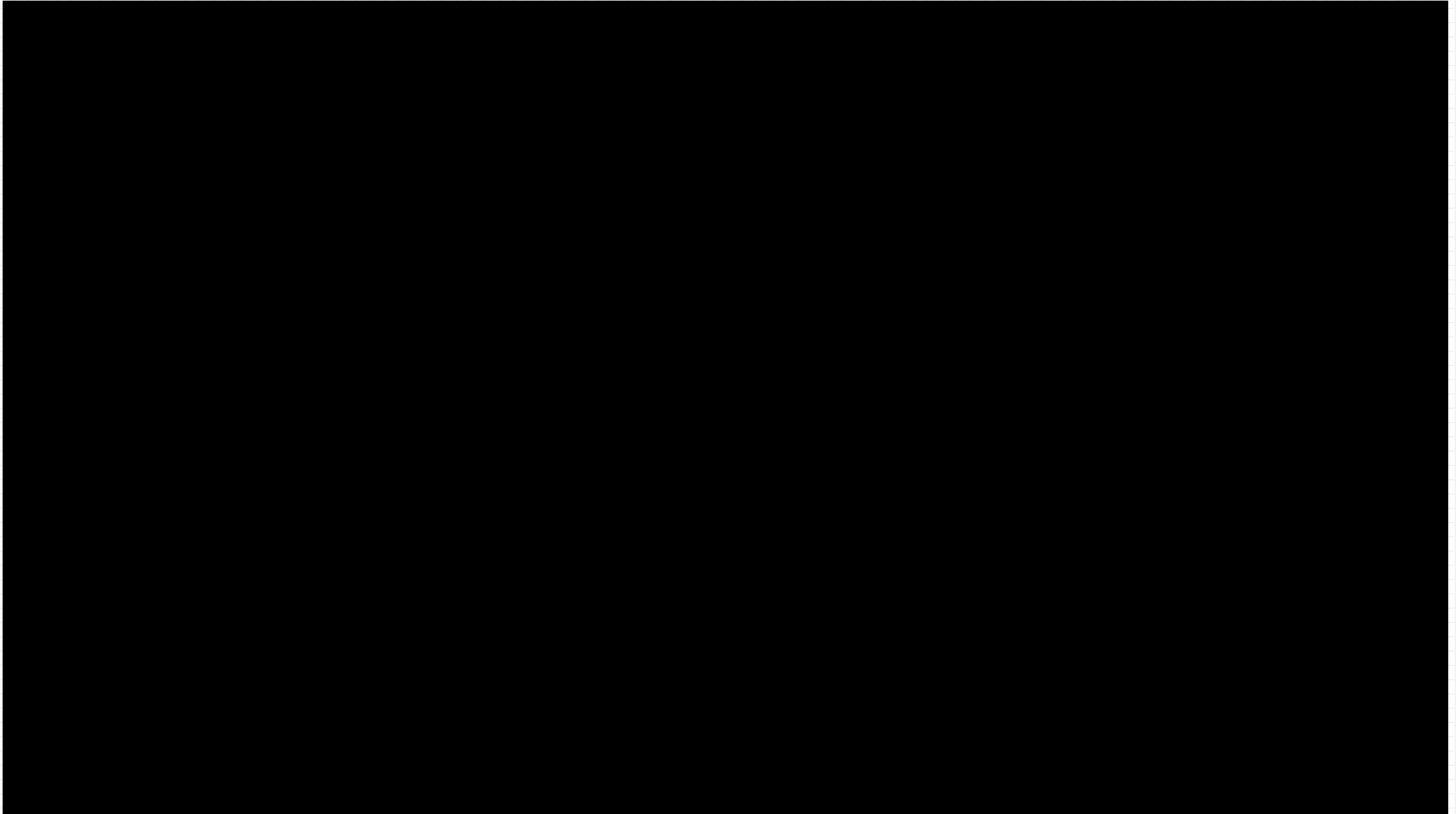


Upcoming CHI submission -> Thermal rendering in VR

Rapid prototyping of flexible circuit



Rapid prototyping with reusing



Zero waste circuit?

* Some parts of this video have been sped up.

DissolvPCB

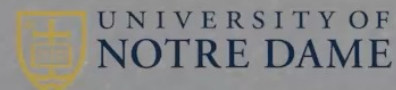
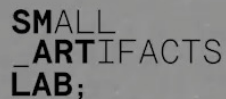
Fully Recyclable 3D-Printed Electronics
with Liquid Metal Conductors and PVA Substrates

Zeyu Yan¹ | SuHwan Hong¹ | Josiah Hester² | Tingyu Cheng³ | Huaishu Peng¹

¹University of Maryland, College Park | ²Georgia Institute of Technology | ³University of Notre Dame

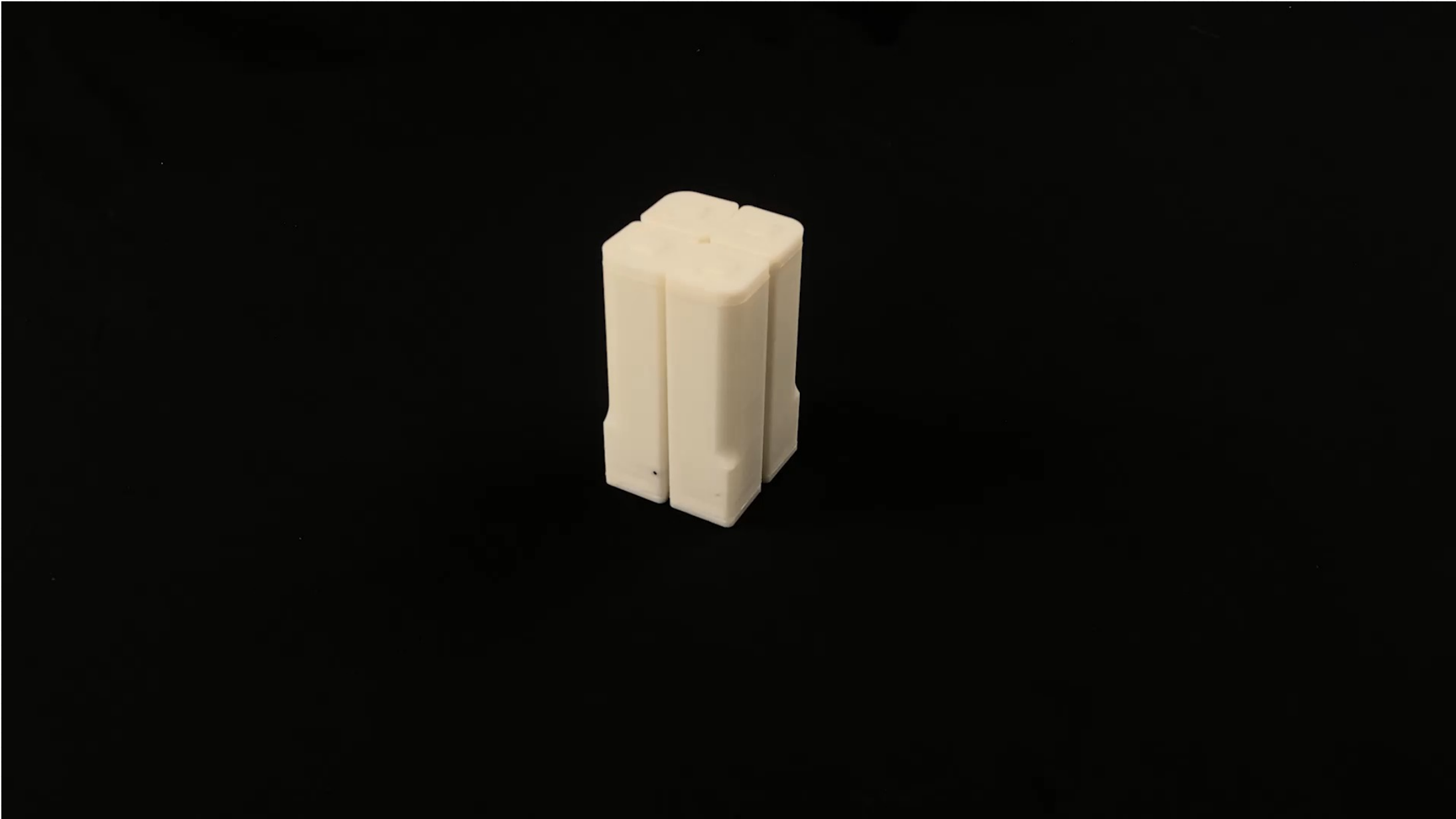


git repo

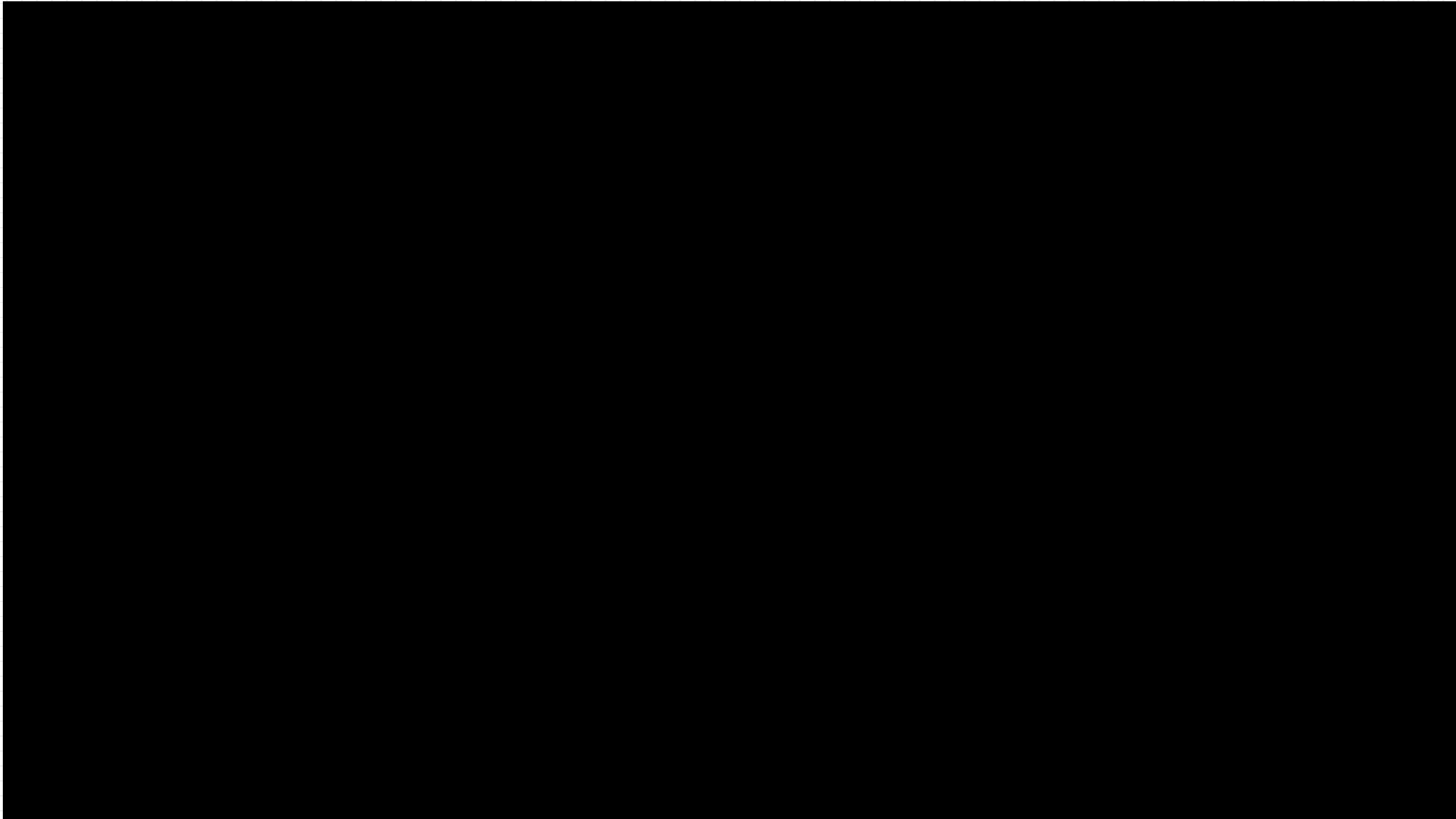


Accessibility

Support blind users to design websites

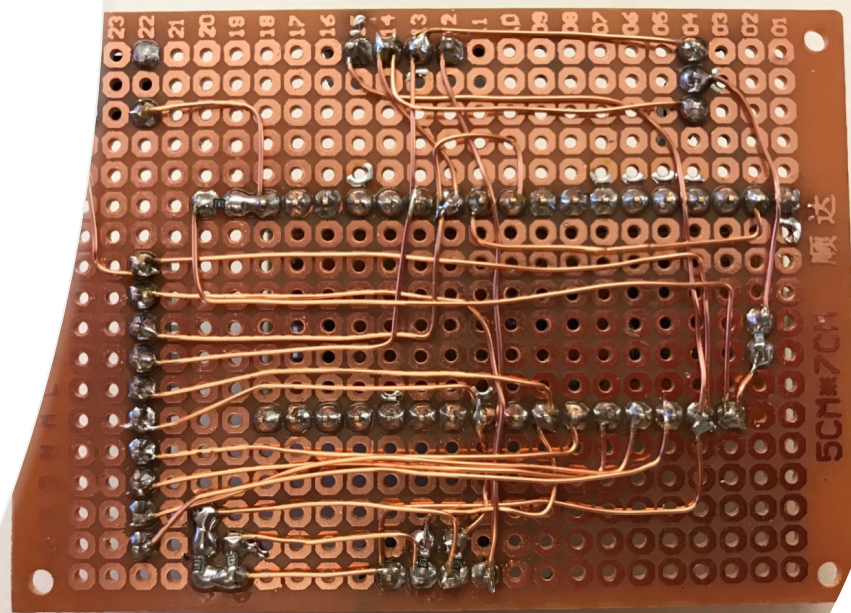
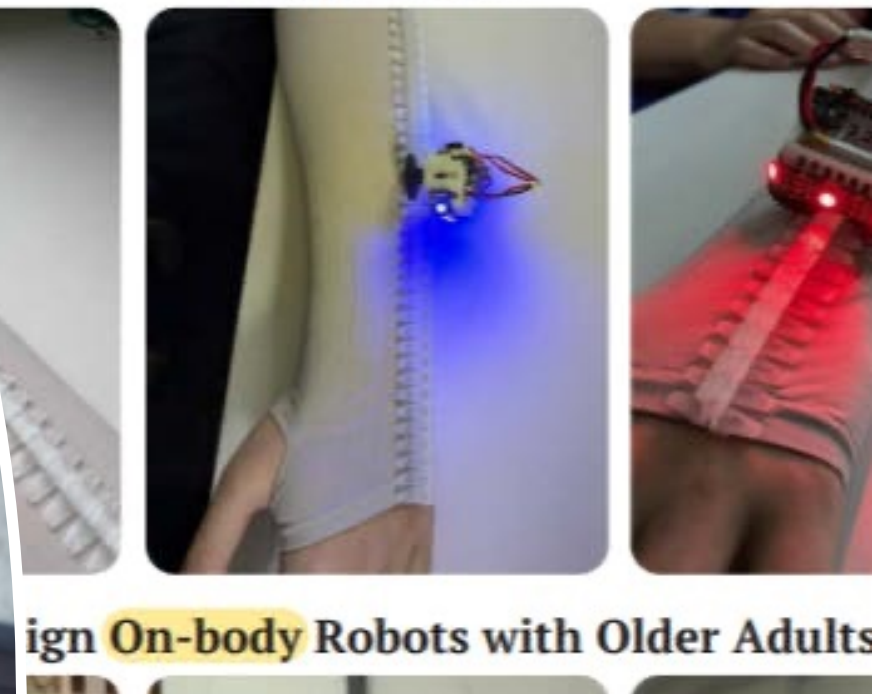


Support blind users to use public touchscreen kiosks



Some of the on-going projects:

If you are interested in doing research with me, drop me a line.



Lab space



IRB 0102

Now about this course

What this course is not about?

It's not (just) about UX

It's not (just) about AI

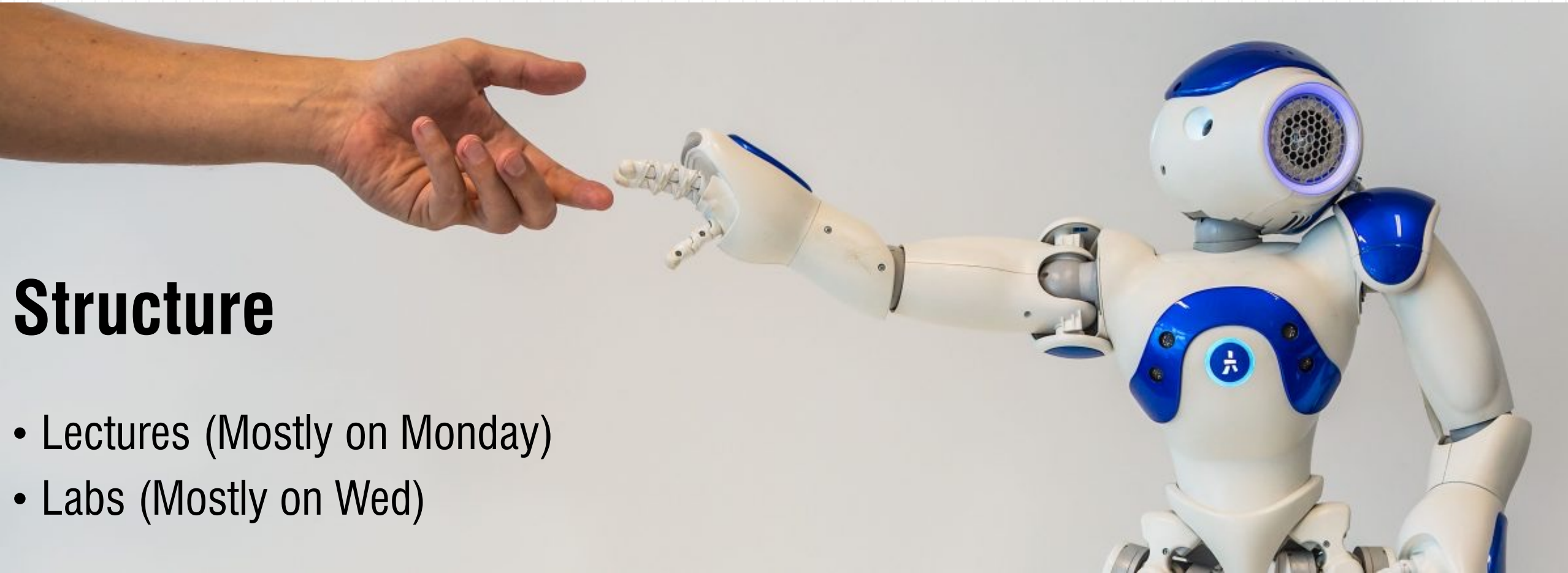
It's not (just) about robot

It's not (just) about user study

Course Overview

HRI | Device Prototyping | Embodied AI



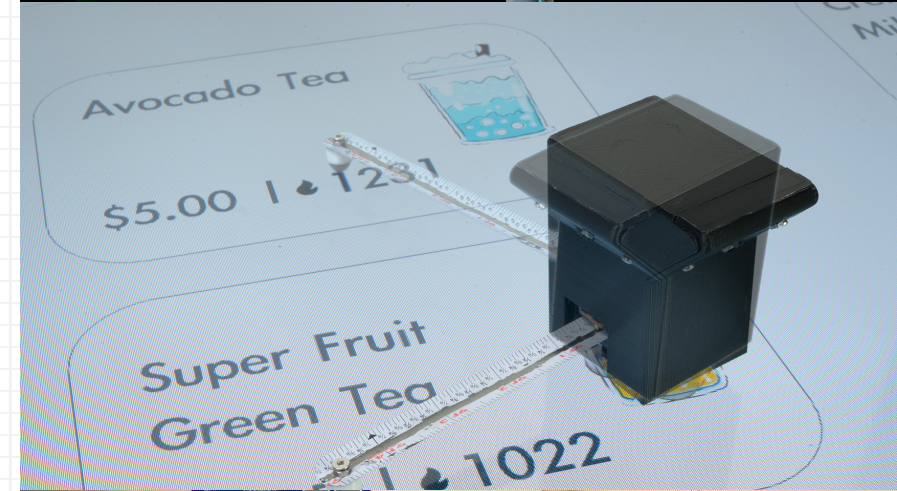
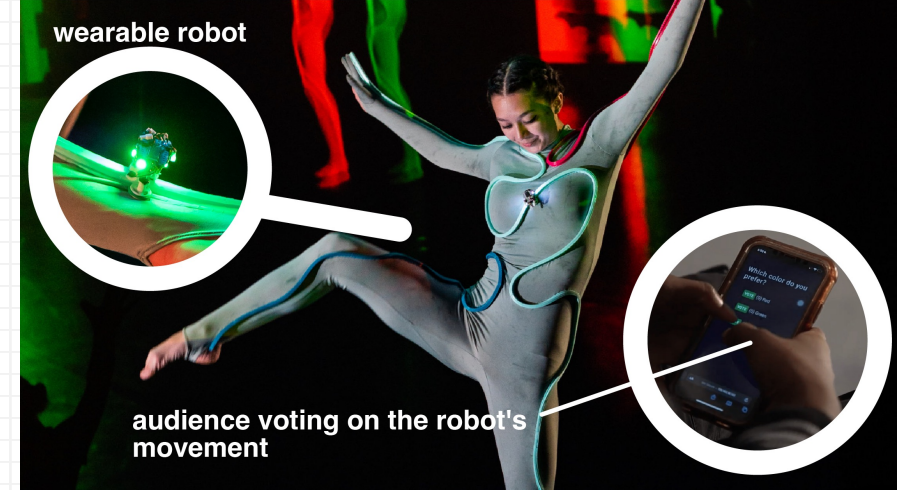


Structure

- Lectures (Mostly on Monday)
- Labs (Mostly on Wed)

Lectures

- Embodied cognition, sensing, planning, learning..
- Empirical research, user studies, interaction paradigm...



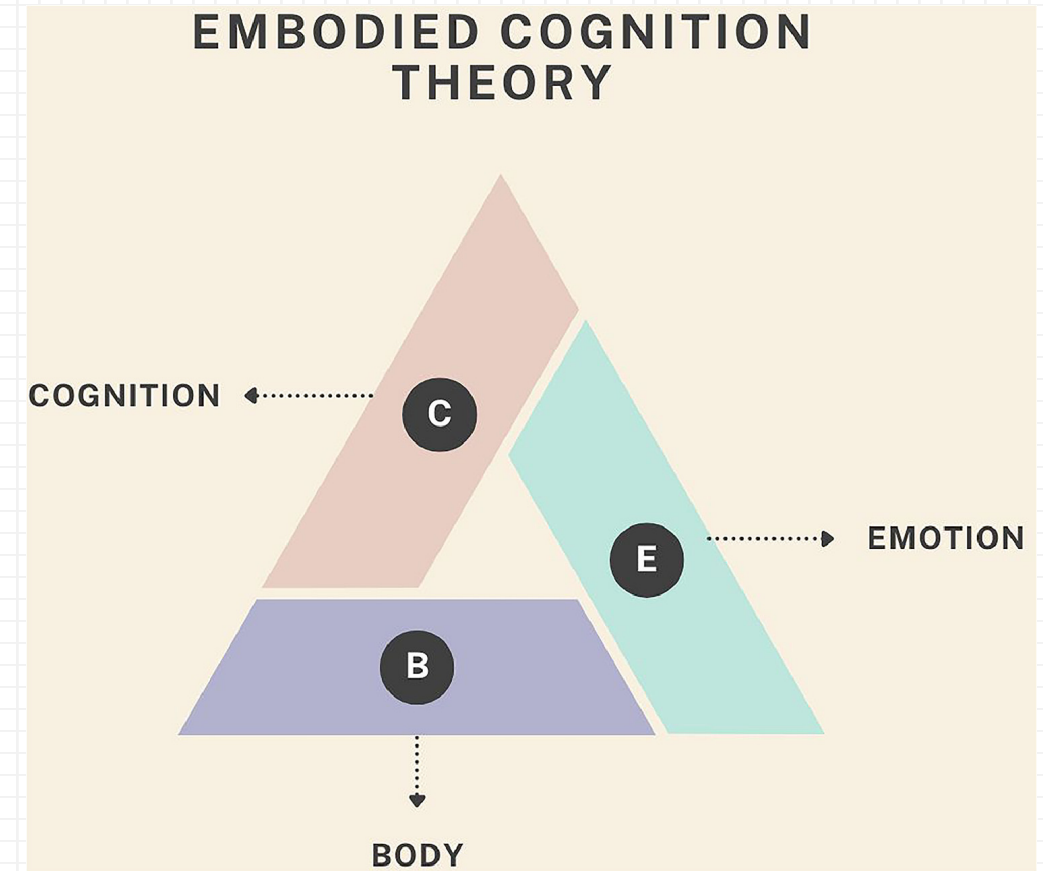
Embodied Cognition

Not only in the “brain” but emerges from the **body’s interaction with the world**

Cognition is grounded in **sensorimotor experience**

Examples:

- A robot’s physical shape influences what tasks it can perform
- Humans rely on gestures and posture to think and communicate



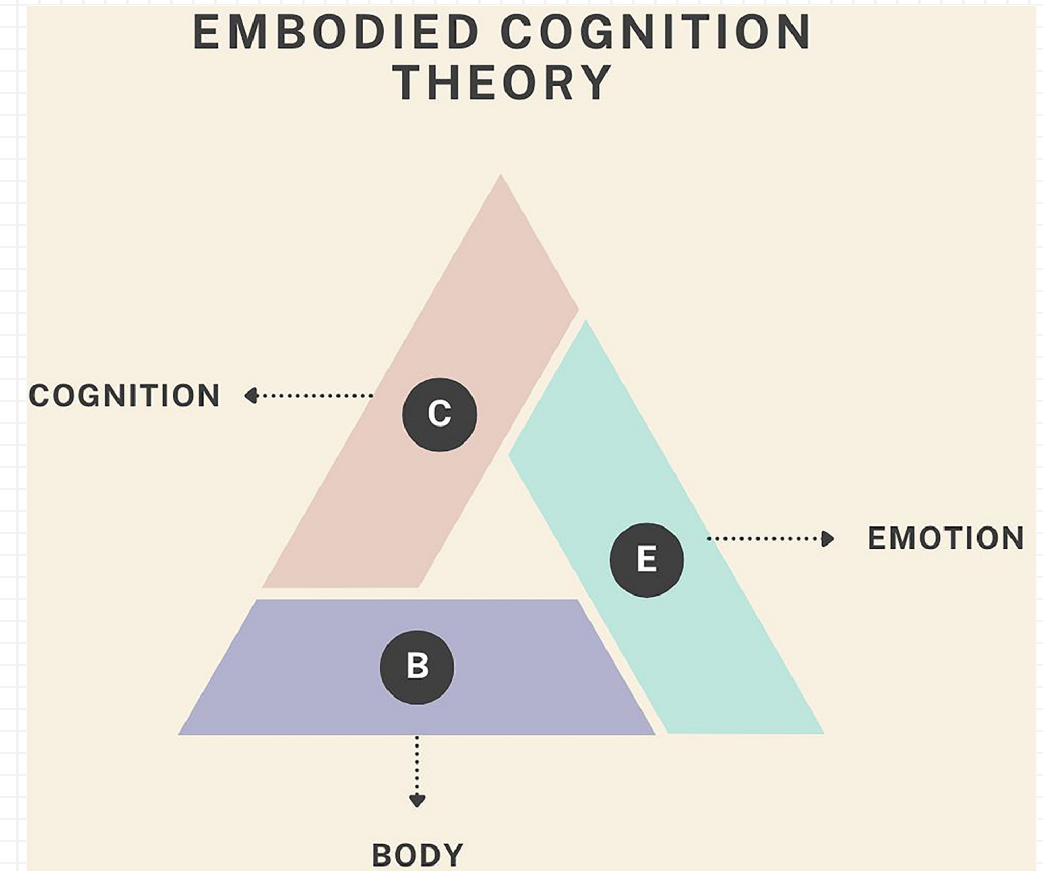
Embodied Cognition

Why important:

Moves AI beyond abstract symbol manipulation
Critical for robots that adapt to real-world environments

You will learn:

How embodiment shapes perception & action
Case studies: passive dynamic walkers, soft robotics
How to design prototypes that “think with their bodies”



Perception, Planning & Learning in Embodied AI

Sensing (Perception): Gathering data about the world (vision, audio, touch, proprioception).

Planning: Sequencing actions to achieve goals under uncertainty.

Learning: Improving performance from experience (supervised, reinforcement, imitation).

These form the **cycle of autonomy**:

Sense → Plan → Act → Learn → Repeat



Perception, Planning & Learning in Embodied AI

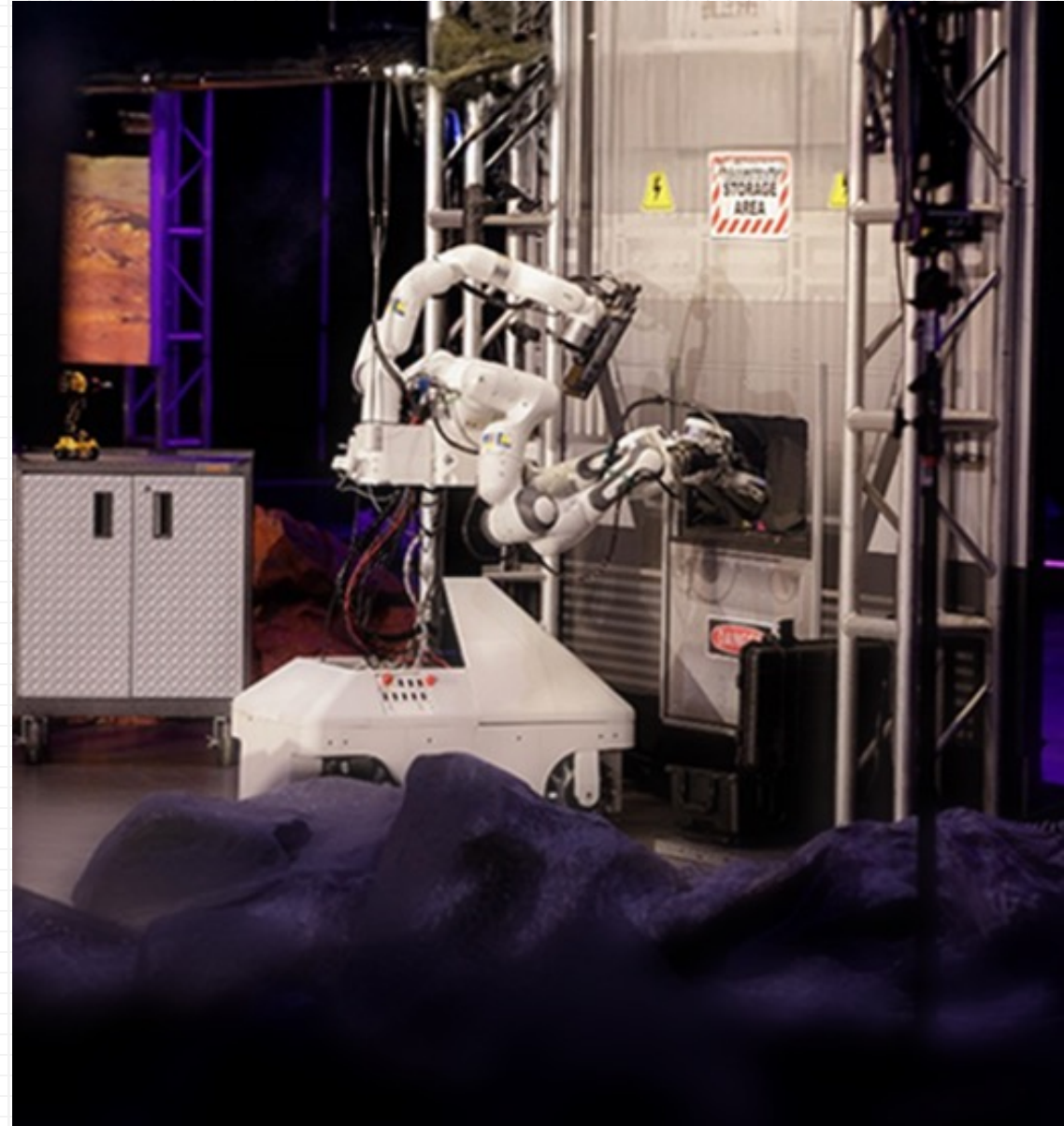
Sensor trade-offs & integration

Planning algorithms (graph search, reactive planning).

Learning methods (RL, supervised/unsupervised for perception).

How perception & planning constrain each other.

How learning reshapes both sensing & planning.



Empirical Research & User Studies

Systematic evaluation of HRI and embodied AI systems

Methods:

- > Experimental design
- > Hypothesis testing
- > Qualitative (interviews, observations) & quantitative (metrics, logs) data



Empirical Research & User Studies

Why important:

Systems succeed/fail based on user trust & usability

Research ethics (IRB, informed consent) ensure responsible design

You will learn:

Basics of study design & data collection

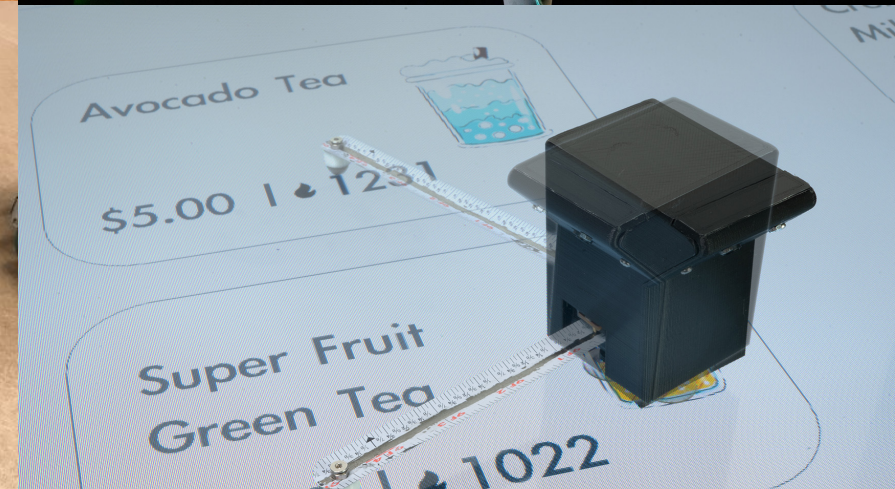
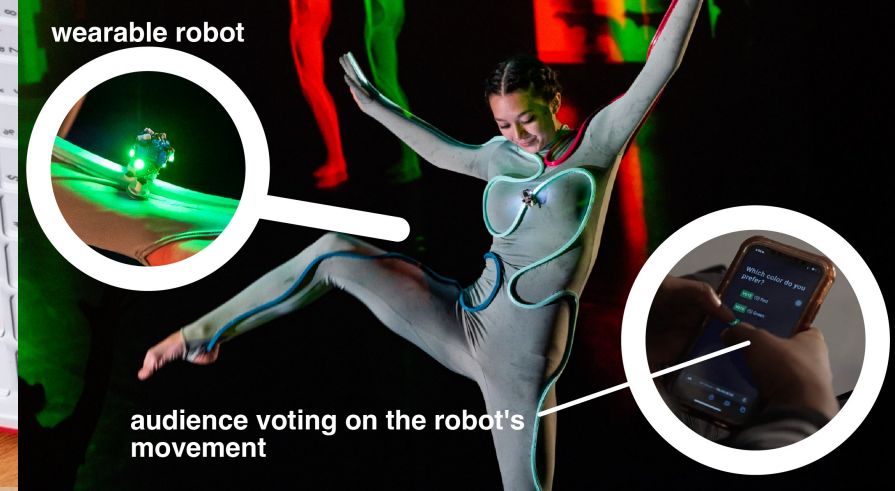
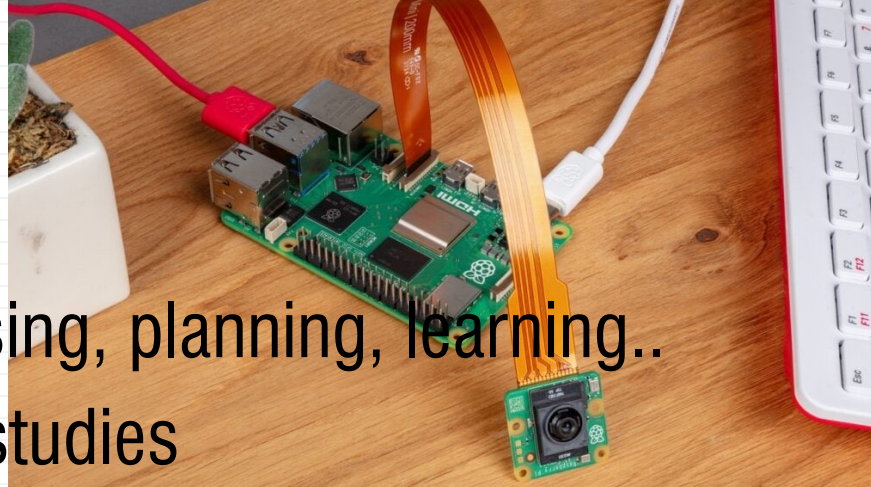
Metrics for HRI (e.g., task efficiency, trust, workload)

How to run pilot studies for prototypes



Lectures

- Embodied cognition, sensing, planning, learning..
- Empirical research, user studies



Labs

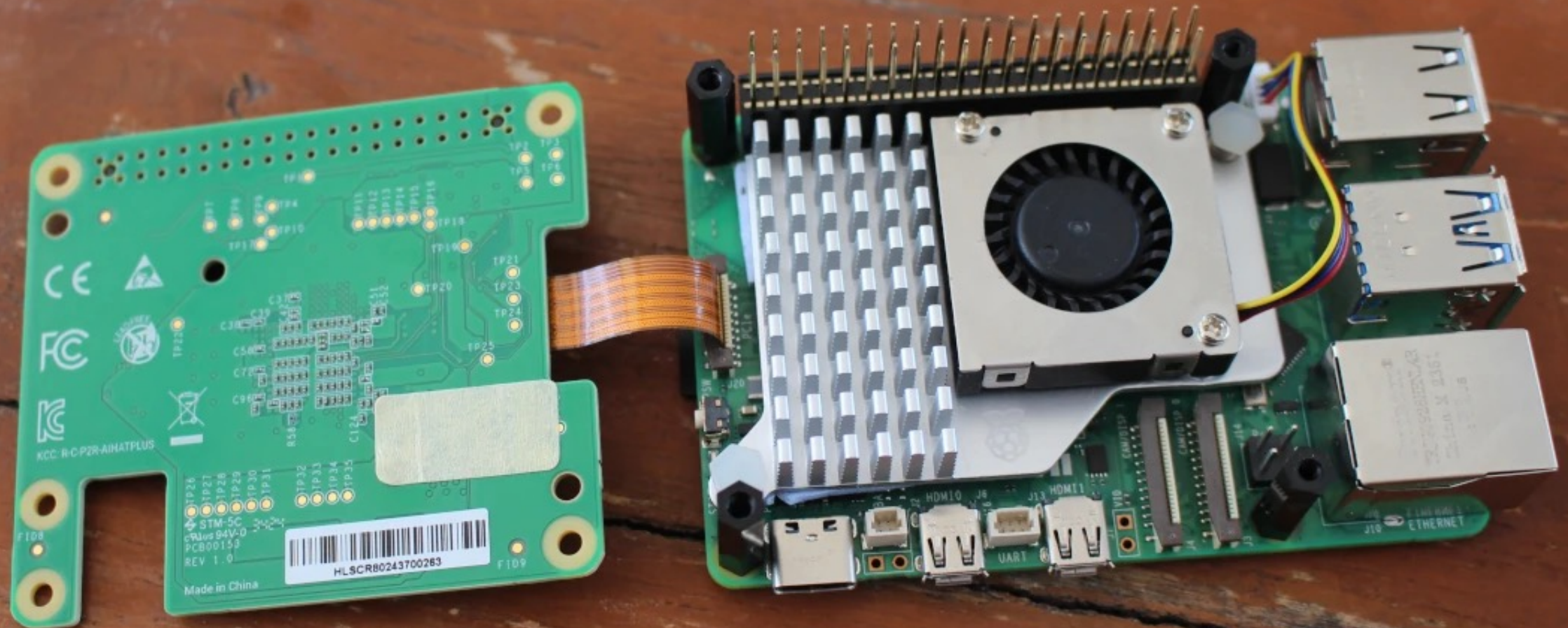
- Hands-on building skills
- Build interactive robots



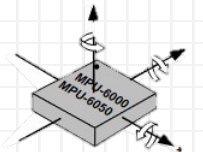
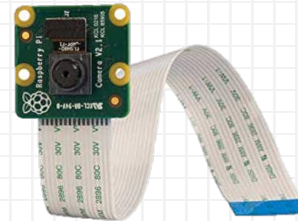
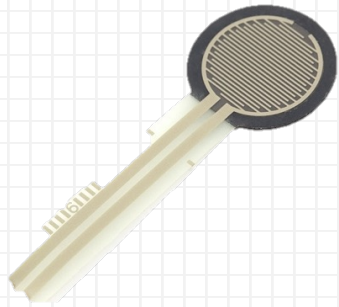
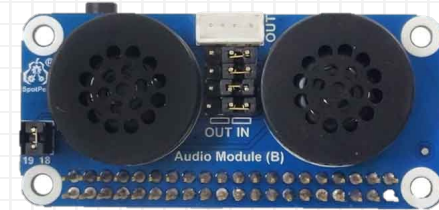
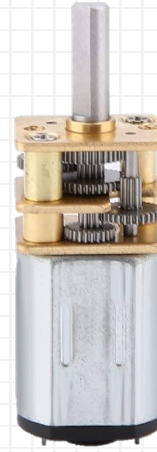
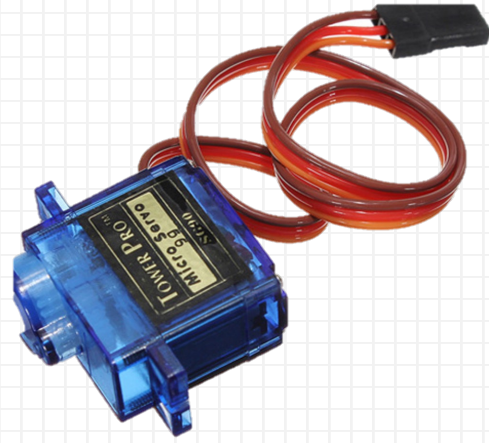
All labs are

Based on a pair of 2, with the following class kit:

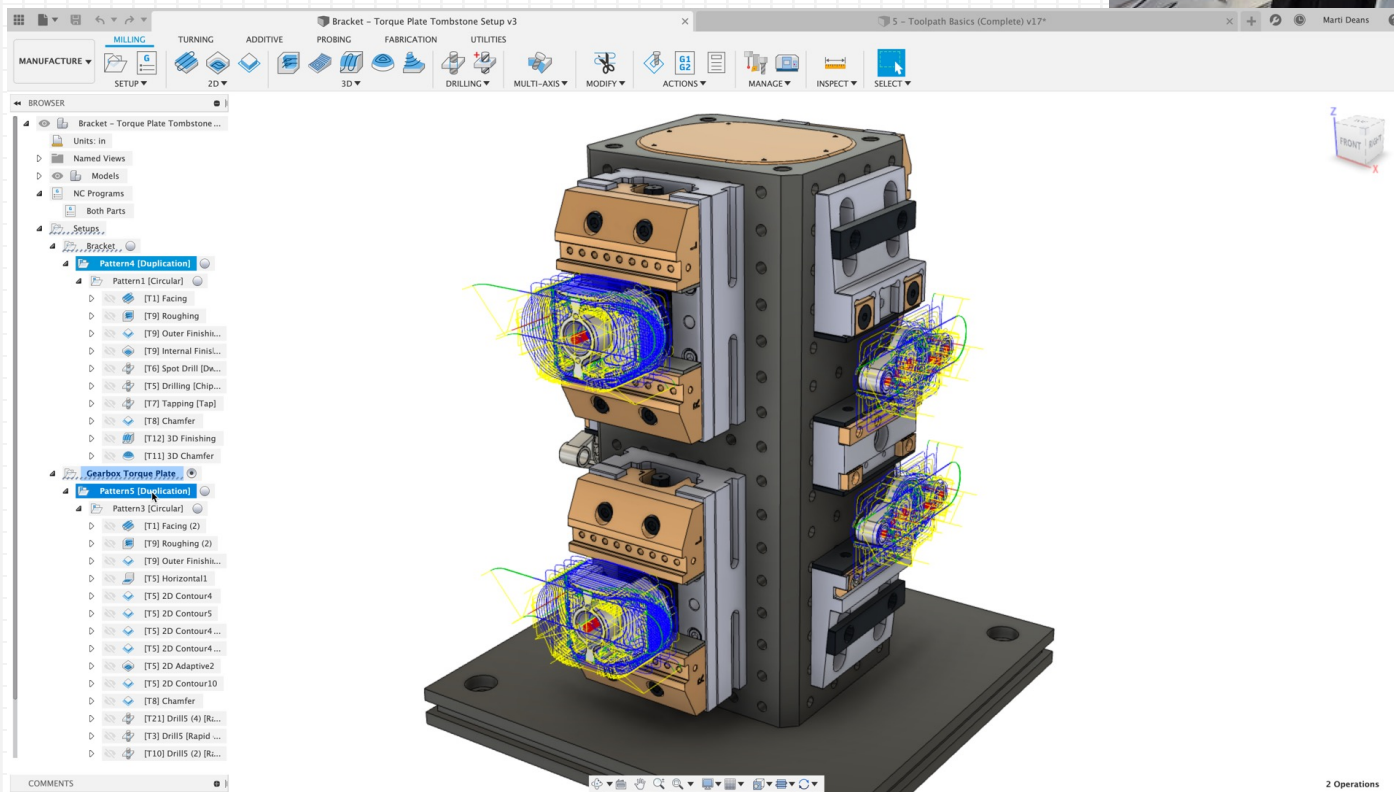
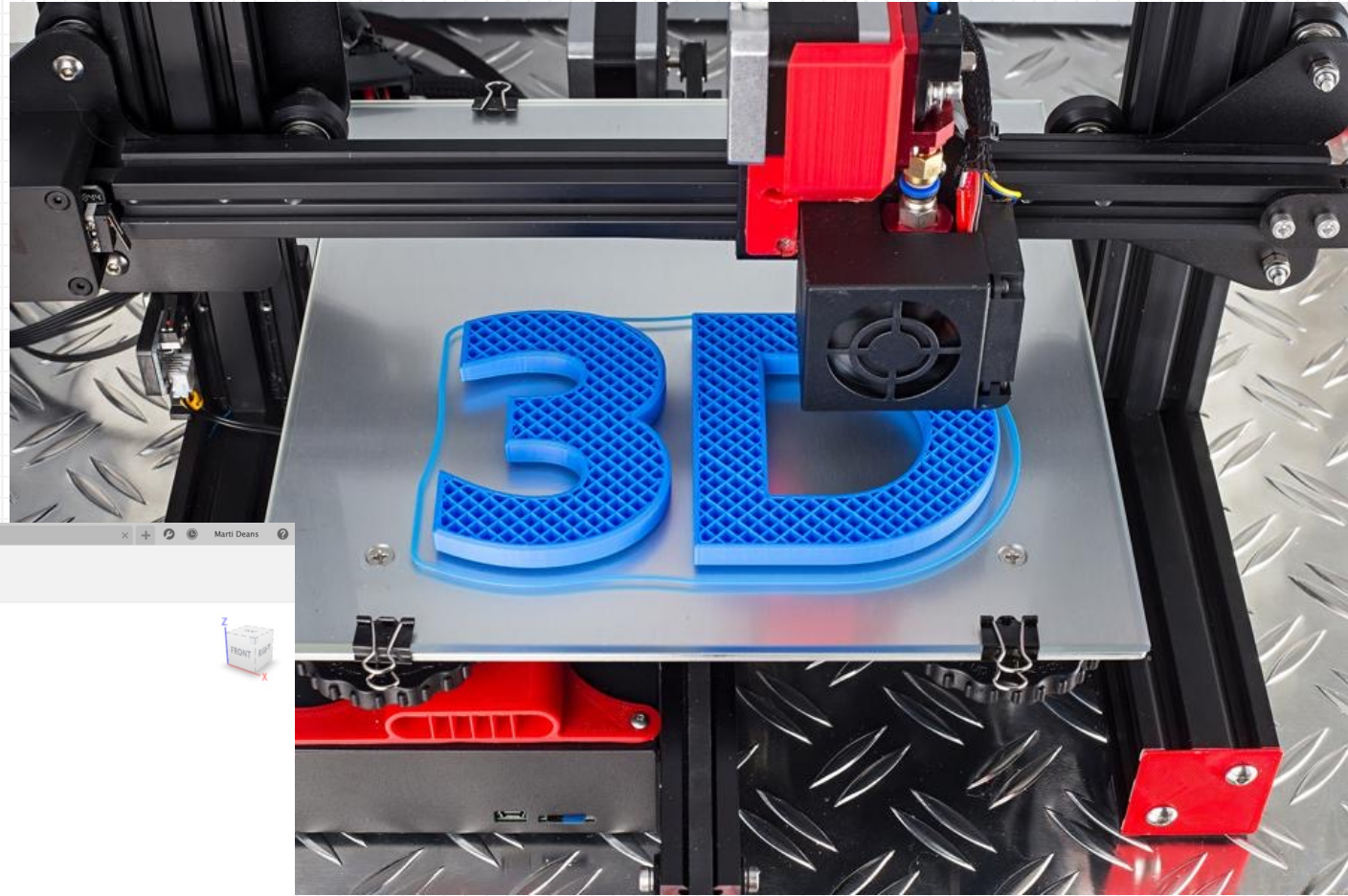
Class Kit



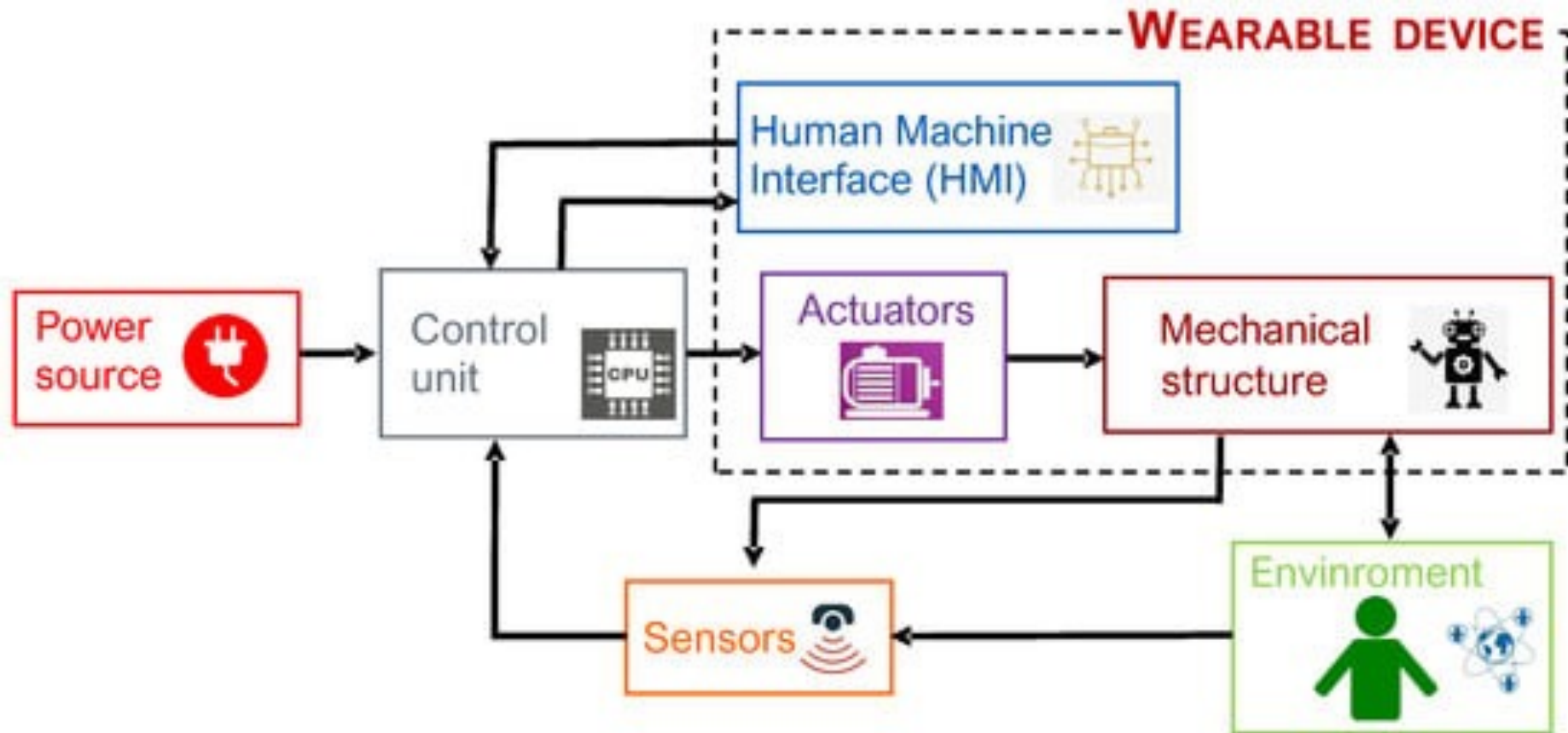
Class Kit



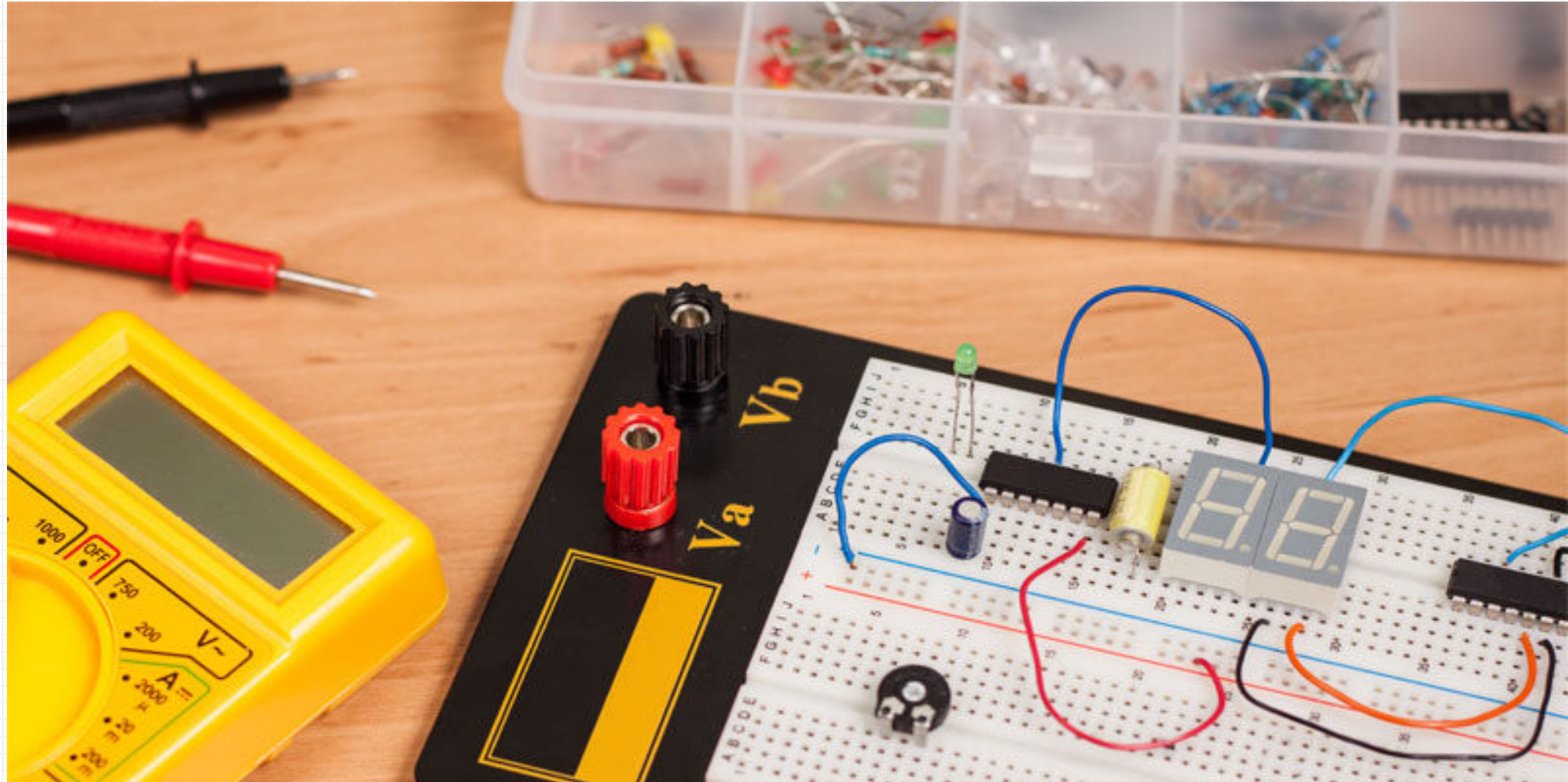
3D Modeling (and of course printing)



Basic sensor + actuation



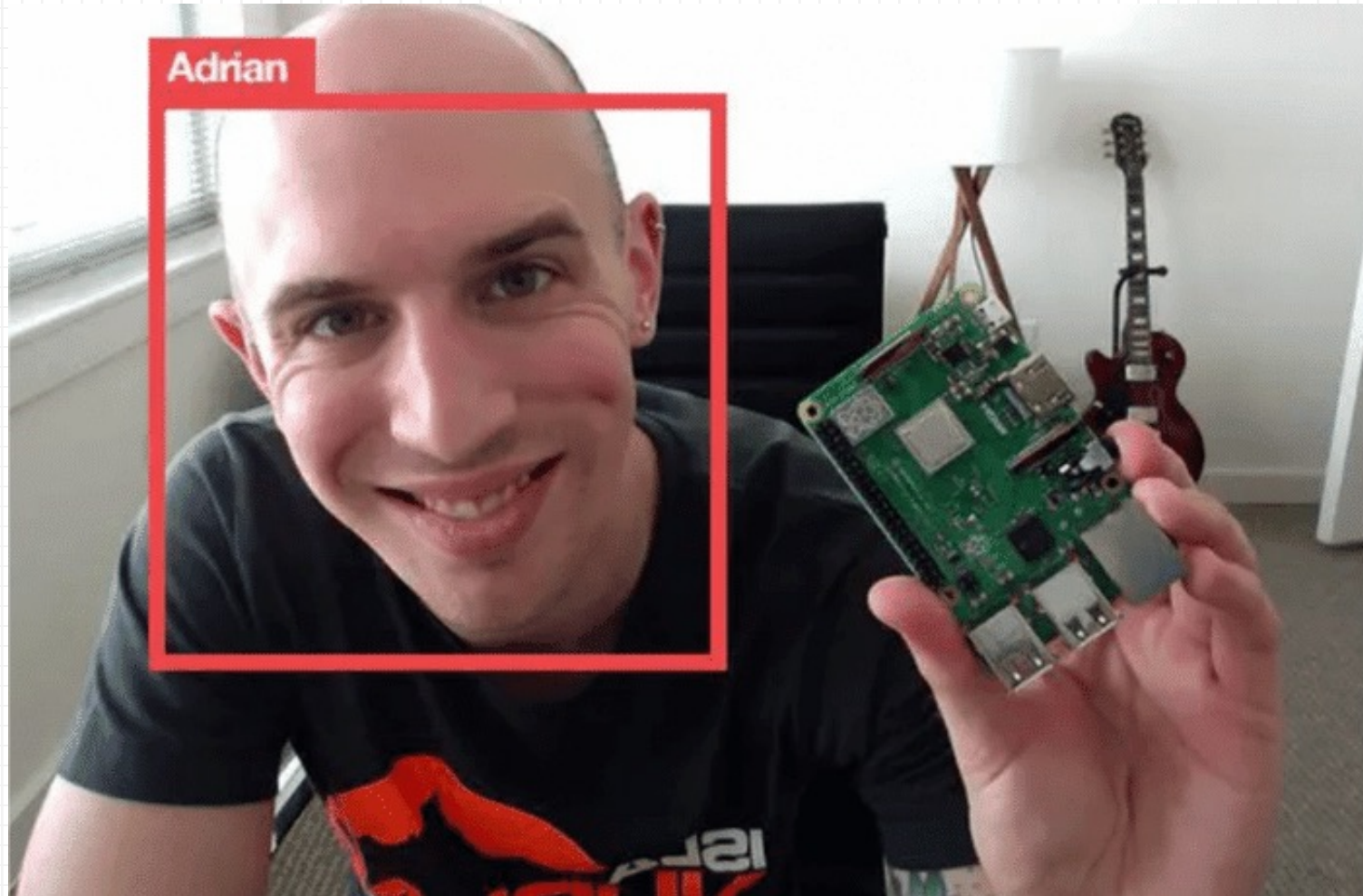
Electronics



ROS2



Detection, classification, display



Semester-long team project

You - a group of 4 - will build

- 1) an interactive robot
- 2) Run user studies with it
- 3) Live demo
- 4) Report

Semester-long team project

Robot requirement:

- a) Has to be a robot (but what is a robot??)
- b) Need to be **interactive**
- c) Not strict replication – **novelty**

Semester-long team project

Study requirement:

- a) Must involve at least 5 human participants.
- b) Study should test or validate some aspect of your robot.
- c) Collect evidence (observations, surveys, or performance data) to support your findings.

Semester-long team project

Deliveries:

- a) Final project report in 1) **UIST | HRI paper format** or
2) **Hackster.IO Online documentation**
- b) Work-in-progress report, can be google doc
- c) **Videos** on YouTube

We will cover basic techniques, but you are likely to **learn many new skills on your own.**

Don't be afraid of making mistakes,
TA and I are here to help

That's all very exciting but...

how much will it cost?

Do I need to pay a material fee?

Material fee

~~We charge you \$130 material fee~~

Material fee

We will provide you **\$130-worth** basic electronic components -> That's the class kit you will return at the end of the semester

We will also create a pool of electronics, sensors, that you can use for your semester-long project

If your team project requires something unique, talk to me.

I will purchase it if it can be reused for future classes.

If all sounds good ->
Declare your

- 1) Lab team (2 per team) by **Mon next week**
- 2) Project group (4 per group) by **Wed next week**

[Home](#)[Course Schedule](#)

Embodied AI & Human–Robot Interaction

Fall 2025 – Senior Undergraduate Course

Welcome!

This new undergraduate course explores how **embodied AI** and **human–robot interaction (HRI)** come together to shape the robots of today and tomorrow. Through lectures, hands-on labs, user study design, and a creative capstone project, you will learn how robots sense, act, learn, and interact with people in meaningful/creative ways.

What to Expect

- **Lectures (Mondays):** Foundations of embodied cognition, robotics, sensing, planning, learning, and interaction design, along with ethics, creativity, and the future of embodiment.
- **Labs (Wednesdays):** Hands-on robotics with Raspberry Pi as the base platform. You will need to wire sensors, controll motors, integrate perception, experiment with ROS2, and build expressive systems.
- **Capstone Project:** In teams, you'll design and build a **novel robot embodiment**. Each capstone will include both a **working prototype** and a **small user study** to test how people respond to your robot, and a final demo in a public showcase.

Tools & Platforms

Each student pair will be provided with a **robotics kit** and access to makerspace resources. You'll work with:

- Sensors: camera, microphone, IMU, tactile sensors.
- Actuators: motors, servos, LEDs, displays, and speakers.
- Software: Python, ROS2, OpenCV, and ML libraries.
- Makerspace tools: 3D printing and craft materials for creative embodiments.

How you will be evaluated

Reading reflections & mini-assignments – 15%

Weekly labs & deliverables – 25%

Checkpoints (proposal + midterm demo) – 20%

Capstone project (final showcase) – 40%

- Functionality
- Creativity & embodiment
- Interaction design & user study
- Presentation

Reading Reflections & Mini-Assignments (15%)

Short write-ups based on (bi-)weekly readings

Connects **theory to practice**

Keeps you engaged with ideas beyond lab work

Graded on completion & thoughtfulness, not length

Weekly Labs & Deliverables (25%)

Hands-on practice with sensors, motors, ROS2, etc.

Small weekly deliverables: logs, screenshots, short reports

Builds technical foundation for your capstone project

Checkpoints (20%)

Checkpoint 1: Project proposal (idea, goals, embodiment)

Checkpoint 2: Midterm demo (prototype, feedback)

Ensures you don't wait until the end

Feedback from peers & instructors to refine your project

Capstone Project (40%)

Team project: design, build, and evaluate a robot

Graded on:

- Functionality
- Creativity & embodiment
- Interaction design & user study (with ≥ 5 participants)
- Presentation & demo & report

Enrollment

We will mainly use ELMS. The class website is a good one-place entry for everything.

ELMS for uploading assignments and for posting discussion questions, e.g., forming class teams.

Resources

Course material will be given to you **2 weeks** after the semester starts

All lectures will be recorded in case you run into difficulties

Office hours:

Wed 2:00-3:00 pm – IRB4126

TA by email appointment

Resources

LAB IRB 0102 – TA resource

IRB Sandbox (1st floor)



Singh Sandbox
Makerspace

sandbox.iri.be.umd.edu

<https://sandbox.iri.be.umd.edu/>

For the rest of this class

Complete this online survey so that I can better know you and know what you would like to learn from this class.